Improving Retinal Blood Vessel Segmentation in U-Net Using Architectural Modifications

Technical Report

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Abstract

Automated segmentation of retinal blood vessels is critical for early detection and monitoring of ocular diseases such as diabetic retinopathy, glaucoma, and hypertension. This study investigates enhancements to the U-Net architecture for retinal vessel segmentation on the FIVES dataset, specifically focusing on Swin U-Net and Attention U-Net with ASPP. We conducted systematic ablation studies to evaluate the impact of simplifying encoder and decoder components, removing skip connections, and altering bottleneck structures. Our experiments showed that removing skip connections significantly degraded performance, while a simplified encoder-decoder architecture reduced overfitting without substantial loss in segmentation quality. We further explored the effect of different loss functions (Dice Loss vs. Focal Tversky Loss) and regularisation techniques (data augmentation and dropout). The link to the implementation is here: https://github.com/ /tree/main

Keywords Retinal Vessel Segmentation, Swin U-Net, U-Net, FIVES

1 Introduction

1.1 Motivation

Retinal fundus imaging provides a non-invasive window into vascular health and is widely used in ophthalmology to diagnose and monitor conditions such as diabetic retinopathy, glaucoma, and hypertensive retinopathy. Accurate segmentation of blood vessels in these images is essential for quantitative analysis, disease tracking, and treatment planning. Manual annotation is time-consuming and prone to inter-observer variability, due to their varying widths, low contrast against the background, and often obscured by pathological features. Thus, automated segmentation is critical for early diagnosis and monitoring of conditions.[1]

1.2 Objectives

The primary goal of this project is to explore the effect of introducing transformer-based networks to improve retinal blood vessel segmentation accuracy on the FIVES dataset by enhancing the U-Net architecture. Initially, we began with a baseline U-Net model and then experimented with various architectural components, including the encoder, decoder, bottleneck, and skip connections. Additionally, we trained and evaluated advanced architectures such as Swin U-Net, and Attention U-Net with Atrous Spatial Pyramid Pooling (ASPP). The results of these experiments were compared. We also analysed the impact of different loss functions, specifically Dice Loss and Focal Tversky Loss.

2 Related Work

2.1 Retinal Vessel Segmentation

Traditional approaches to retinal vessel segmentation relied on hand-crafted features and morphological operations. Recent deep learning methods have significantly outperformed classical techniques. Various U-Net modifications have been proposed, including attention mechanisms, residual connections, and multi-scale feature fusion. However, challenges remain in detecting thin vessels and handling class imbalance (vessels occupy only 10-15% of image pixels).[2]

2.2 U-Net for Medical Image Segmentation

U-Net, introduced by Ronneberger et al. (2015), has become the de facto standard for biomedical image segmentation due to its encoder-decoder architecture with skip connections. The encoder progressively reduces spatial resolution while increasing feature depth, capturing hierarchical representations. The decoder upsamples feature reconstructing pixel-wise predictions, with skip connections preserving fine spatial details lost during downsampling. This design is particularly effective for tasks requiring precise localisation, such as vessel segmentation.[3]

2.3 Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation

This paper presents Swin U-Net, a U-Net–like pure transformer model for medical image segmentation. The key advantage of this architecture is its ability to focus on the global context of an image, allowing pixels to communicate with each other even when they are far apart. So Transformer blocks can directly establish dependencies between similar features.

The input images are divided into token patches and processed by a Swin Transformer-based U-shaped encoder-decoder with skip connections. The encoder uses shifted windows to capture contextual information, while the decoder mirrors this structure. In the end, the patches are expanded to recover spatial resolutions. The model was tested on multi-organ and cardiac segmentation tasks.[4]

2.4 What Makes Our Approach Different?

Unlike previous studies that focus on adding only complex modules to the U-Net, our approach systematically evaluates the necessity of each architectural component. We also developed our own version of Swin U-Net according to *Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation* article and source code. We demonstrate that simplifying the network can reduce overfitting while maintaining segmentation quality. Furthermore, we provide a comprehensive comparison of loss functions and regularisation strategies specifically for retinal vessel segmentation, establishing a strong baseline for future Transformer integration.

3 Methods

3.1 Dataset

The FIVES (Fundus Images for Vessel Segmentation) dataset consists of high-resolution retinal fundus images specifically designed for vessel segmentation tasks. The dataset includes a total of 800 images, each with a resolution of 2048×2048 pixels (RGB). In the official paper, 600 images are provided for training and 200 for testing. For our experiments, we further subdivided the original data to get a validation set used to monitor training, resulting in 480 training images, 130 validation images, and 200 test images. The ground truth consists of binary vessel masks that have been annotated by experts.[1]

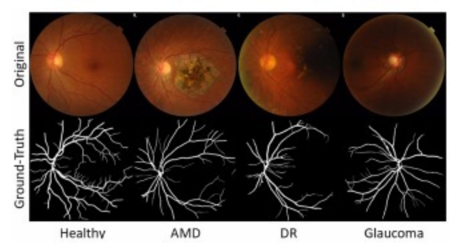

Figure 1: The FIVES (Fundus Images for Vessel Segmentation) dataset contains high-resolution retinal fundus images[1].

Data Preprocessing: To reduce computational costs, we created **smaller versions of the images** with resolutions of 512×512 in both RGB and grayscale formats. In the training set, we also implemented in some experiments **augmentation techniques**, which included horizontal and vertical flips, 90-degree rotations, brightness adjustments, and normalisation to prevent overfitting.

3.2 Network Architectures

3.2.1 Baseline U-Net Architecture

Our baseline U-Net follows the classic structure. Starting with **encoder**, where the model includes four downsampling stages with **DoubleConv** blocks, which consist of two consecutive 3×3 convolutions followed by **Batch Normalisation** and **ReLU** activation. The channel depths are as follows: [32, 64, 128, 256] . At the lowest resolution is the **bottleneck**, which consists of a **DoubleConv** block where input has 256 channels and output has 512 channels. Then, the decoder contains four upsampling stages that utilise **transposed convolutions**. The **decoder** also contains skip connections to concatenate with encoder features, followed by **DoubleConv** blocks.

The **difficulty** is that basic U-Nets struggle with detecting small blood vessels due to the variation in vessel width, contrast, and noise.

3.2.2 Swin U-Net

The main goal of Swin U-Net is to learn deep features of the data and efficiently capture long-range image context. The network architecture has a similar structure to the baseline U-Net, but its blocks differ as they are composed of **(by default double) Swin Transformer Blocks**. Initially, each image is divided into non-overlapping patches (**Linear Embedding**), which serve as tokens that are input into the **Swin encoder** (by default 4 stages). Then, in the middle of the architecture, there is a **Swin Transformer Block bottleneck**. To upsample the features from the encoder, the data are fused in the **Swin decoder** (by default 4 stages). In the final upsampling layer (**Path Expanding**), the tokenised patches are restored to the original image with the output segmentation map.

The Swin Transformer Block includes **Layer Normalisation**, followed by a **window-based multi-head self-attention (W-MSA)** module and a **shifted window-based multi-head self-attention (SW-MSA)** module. Followed by another **Layer Normalisation**. Additionally, the block has a **2-layer Multi-Layer Perceptron (MLP)** with **GELU** activation, along with **Dropout** applied to both the attention and **MLP** components.[4]

3.2.3 Attention U-Net with Astrous Spatial Pyramid Pooling (ASPP)

The Attention U-Net is an improved version of the basic U-Net by adding **Attention Gates** in the skip connections. The purpose of this modification is to retain and extract deeper contextual information (relevant features) from the data through these skip connections.

The Attention Gate consist of 1×1 **Convolution Layers** and **RELU** activations. Our approach differs from the conventional Attention Gate by incorporating **Instance Normalisation**, which helps focus more effectively on the content while applying the new style. This addition aims to reduce false positive predictions, minimise noise, and improve the detection of tiny vessels.[5]

ASPP is used in **directly after the bottleneck**, at the deepest resolution. It is crucial to emphasise the various shapes of regions, which is essential for accurately defining the width of vessels. In our implementation, we used ASPP module from the PyTorch library DevLabV3, with dilation rates [6, 12, 18]. The **ASPP** module consists of **convolutional layers** with varying spatial dimensions and dilation rates, followed by **image pooling**. Finally, the data are concatenated with a 1×1 **convolution layer**.[6]

3.3 Loss Functions

We compared two loss functions to address class imbalance:

Dice Loss: $L_{\text{Dice}} = 1 - \frac{2|P \cap G| + \epsilon}{|P| + |G| + \epsilon}$

where P is the predicted vessel mask, G is the ground truth, and $\in$ is a smoothing term. Dice Loss directly optimizes the Dice coefficient (F1-score), treating precision and recall equally. It is robust to class imbalance as it measures overlap rather than pixel-wise accuracy.[2] [7]

Focal Tversky Loss:

$$L_{\text{FT}} = (1 - TI)^{\gamma} \qquad TI = \frac{TP + \epsilon}{TP + \alpha FP + \beta FN + \epsilon}$$

where TP, FP, FN are true positives, false positives, and false negatives, respectively. Parameters α and β control the trade-off between precision and recall, while γ adjusts focus on hard examples. We used α **= 0.3**, β **= 0.7** to emphasize recall (detecting all vessels, even at the cost of some false positives), and γ **= 1.333**.

Focal Tversky Loss is particularly effective for detecting small structures like thin vessels, as it can be tuned to prioritize sensitivity over specificity.[2] [8]

3.4 Evaluation Metrics

We evaluated model performance using three metrics:

DICE Coefficient (F1-Score): $\text{DICE} = \frac{2TP}{2TP + FP + FN}$

Measures overlap between predicted and ground truth masks, balancing precision, and recall.[2] [8]

Focal Tversky Index: As defined in the loss function section, it provides asymmetric weighting of false positives and false negatives.[2] [8]

Area Under the ROC Curve (AUC): Evaluates classifier performance across all decision thresholds, measuring the ability to distinguish vessels from non-vessel pixels.[2] [8]

3.5 Experiments

We mostly trained with **epochs 30 for U-Net and 40 for Swin U-Net**, using **early stopping** to monitor either the **Dice loss** (DL) or the **FocalTversky loss** (FTL). The optimiser we used was **Adam**. The output activation function was **sigmoid**, and we set the **initial learning rate to 0.0001**, utilising a **cosine learning rate scheduler** that had a **minimum learning rate of 0.000001**. In some cases, we also applied **dropout** with a rate of **0.1** in those experiments that used augmentation. Validation performance was also monitored. Training curves and metrics were visualised in real-time using **TensorBoard**.

We experimented on two devices. The first setup was equipped with an NVIDIA 3060 6GB VRAM and an AMD Ryzen 7 5800HS. The second device was a Mac running MacOS Sequoia 15.3.1 on an Apple M1 chip with 16GB of RAM.

We begin our experiment with a standard baseline U-Net model. Subsequently, we analysed and modified various components of the U-Net to assess their impact (described in our internal notes, due to page limit). From this results we created the designed **simplified Encoder-Decoder U-Net**, where **DoubleConv** blocks are replaced with **SingleConv** blocks at the encoder and decoder, improved performance. Therefore, the baseline U-Net indicated overfitting. Thus, **we included data augmentation and dropout** (represented by "reg" in the experiment names). To address the imbalance of the data, we utilise the **Focal Tversky loss**. In the next experiments, we turned our attention to the **Swin U-Net** architecture and conducted a series of tests to explore its behaviour and performance. Where we tested different patch sizes, window sizes, embedding dimension, as well as with a small convolutional pre-processing block (conv_stem). Finally, we tested the **Attention U-Net with ASPP**.

Table 1: Summary of U-Net, Swin U-Net, and ASPP experiments.

Model	Experiment	Dice (Val) ↑	Focal Tversky (Val) ↑	AUC (Val) ↑	Key configuration / comment
Baseline vs. Simple Encoder–Decoder (Dice loss, no reg)					
Baseline	exp003_unet_components_01_baseline	**0.8683**	0.6565	**0.9772**	DoubleConv encoder/decoder, skip connections
Simple Encoder-Decoder	exp003_unet_components_13_simple_encoder_decoder	0.8521	**0.8269**	0.9736	SingleConv encoder/decoder, DoubleConv bottleneck, skip connections
Baseline vs. Simple Encoder–Decoder (Focal Tversky loss, reg)					
Baseline	exp006_focal_red_01_baseline	0.8272	0.8611	0.9770	baseline U-Net, data augmentation, dropout
Simple Encoder-Decoder	exp006_focal_red_11_simple_encoder_decoder	**0.8320**	**0.9061**	**0.9802**	simple encoder–decoder, augmentation, dropout
Swin U-Net (Focal Tversky loss, Dice loss, reg)					
Swin U-Net - DL	exp008_tinyswin_reg_dice_patch2_embeddingDim64	**0.8051**	0.8790	0.9745	patch_size = 2, embedding_dim = 64, window_size = 4, conv_stem = false
Swin U-Net - FTL	exp008_tinyswin_reg_focal_patch2_embeddingDim64	0.7748	0.8822	0.9768	patch_size = 2, embedding_dim = 64, window_size = 4, conv_stem = false
Swin U-Net - DL	exp008_tinyswin_reg_dice_patch4_embeddingDim128	0.7453	0.8311	0.9498	patch_size = 4, embedding_dim = 128, window_size = 4, conv_stem = false
Swin U-Net - FTL	exp008_tinyswin_reg_focal_patch4_embeddingDim128	0.7055	0.8311	0.9509	patch_size = 4, embedding_dim = 128, window_size = 4, conv_stem = false
Swin U-Net - DL	exp008_tinyswin_reg_dice_deep_noConvstem	0.7521	0.8314	0.9656	patch_size = 2, embedding_dim = 128, window_size = 4, conv_stem = false
Swin U-Net -FTL	exp008_tinyswin_reg_focal_deep_noConvstem	0.7828	0.8854	**0.9771**	patch_size = 2, embedding_dim = 128, window_size = 4, conv_stem = false
Swin U-Net - DL	exp008_tinyswin_reg_dice_deep_convstem	0.7837	0.8446	0.9429	patch_size = 2, embedding_dim = 128, window_size = 4, conv_stem = true
Swin U-Net - FTL	exp008_tinyswin_reg_focal_deep_convstem	0.7703	0.8638	0.8638	patch_size = 2, embedding_dim = 12, window_size = 4, conv_stem = true
Swin U-Net - FTL	exp009_tinyswin_reg_focal_window2_convstem	0.8009	**0.88885**	0.8965	patch_size = 2, embedding_dim = 64, window_size = 2, conv_stem = true
Attention U-Net with ASPP (Focal Tversky loss, Dice loss, reg)					
Attention U-Net with ASPP	exp001_aspp_att_unet	**0.8442**	**0.8757**	**0.9299**	Attention Gate in skip connections, ASPP, augmentation, dropout

4 Results

The quantitative results of our experiments are summarized in Table 1. Our Baseline U-Net achieved the highest Dice score (0.8683) when trained without regularization, but suffered from a poor Focal Tversky score (0.6565), indicating a tendency to miss finer vascular structures. The introduction of the simplified encoder-decoder architecture proved highly effective. When combined with Focal Tversky Loss and regularization (augmentation and dropout), the simplified model achieved the best overall balance of metrics: a Dice score of 0.8320, the highest Focal Tversky score of 0.9061, and the highest AUC of 0.9802. This confirms our hypothesis that reducing model complexity can mitigate overfitting on datasets of this size. Conversely, the Swin U-Net architectures generally struggled to outperform the convolutional baselines. The best performing Swin configuration (Patch size 2, Embedding Dim 128, window size 2 and convolutional stem) achieved a Dice score of 0.8009 and a Focal Tversky score of 0.8888. While competitive in this two metrics, it did not surpass the simplified U-Net, likely due to the lack of large-scale pre-training which is typically required for transformer-based models to converge optimally, which can also result in the low AUC value obtained. The Attention U-Net with ASPP showed solid performance (Dice 0.8442, FT 0.8657), serving as a robust middle-ground.

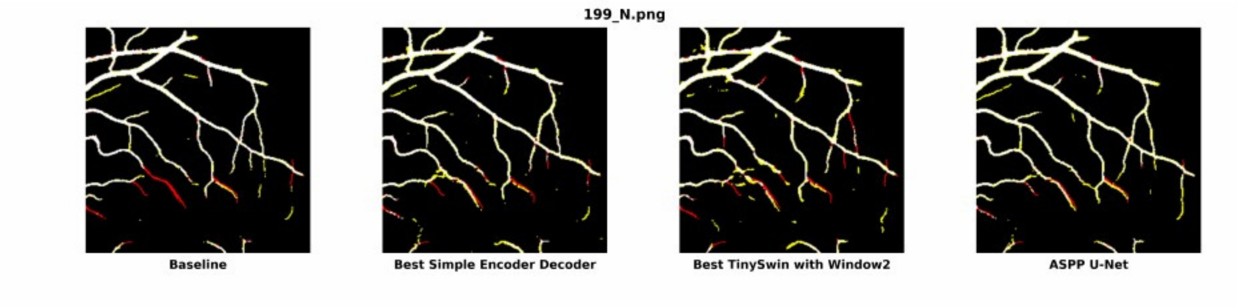

Figure 2: Qualitative comparison between Baseline, Simple Encoder-Decoder, Swin U-Net, Attention U-Net with ASPP

Based on the qualitative comparison in Figure 2 and similar visualizations across the experiments, we observed that the Simplified Encoder-Decoder U-Net yielded a slight improvement in the continuity and detection of thin vessels compared to the Baseline U-Net prediction. Conversely, the Swin U-Net configurations often produced a slightly noisier prediction, while the Attention U-Net with ASPP offered a segmentation performance very similar to the simplified version, but with more continuous vessel detection.

4.1 Comparison of our top-performing models with similar studies.

Based on all the experiments executed, the Baseline U-Net and the Simple Decoder-Encoder U-Net achieved the best results. The Baseline U-Net performed at its best according to the Dice Loss metric, while the Simple Decoder-Encoder U-Net excelled using the Focal Tversky Loss. We compared these models with related work in Table 2.

Table 2: Comparison of the best models with related work

Model	Loss function	Dice (Val) ↑	Focal Tversky (Val) ↑	AUC (Val) ↑
U-Net	Dice	0.8683	0.6565	0.9772
Simple Encoder-Decoder	Focal Tversky loss	0.8320	0.9061	**0.9802**
UGS-M3F[7]	DiceBCE	**0.9848**	–	0.9843
FR-UNet[8]	Dice	0.9004	–	–
UNet[8]	Dice	0.8987	–	–
MA-Net[8]	Dice	0.8988	–	–

We did not outperform the models from related work. The leading model, UGS-M3F, is basically an improved retinal vessel segmenter that mixes the strengths of CNNs and Swin Transformers. The paper's propose solution is effective to detect tiny capillaries and big vessels. The result (Dice score: 0.9848) speaks for itself.

In another study *Benchmarking Retinal Blood Vessel Segmentation Models for Cross-Dataset and Cross-Disease Generalization*, the authors explored various enhancements of baseline U-Net architecture, including FR-Unet (Full-Resolution U-Net) and MA-Net (Multi-Attention U-Net). Despite the differences in architectural complexity, the performance of these models were similar, almost identical. This suggests that when we have enough reliable training data, a simple U-Net can achieve comparable accuracy to more complex architectures while being cheaper to train. Similarly, we observed a related outcome in our case, but with slightly lower accuracy.

In the reviewed related works, the authors explored various loss functions beyond Dice loss due to class imbalance present in the FIVES dataset. They used mostly combination of Dice and Binary Cross-Entropy loss (BCE), while also working with a resolution of 512×512 resolution. It might be beneficial to use the DiceBCE loss in our work as well in the future.

5 Discussion

5.1 Architectural Insights

Our internal ablation studies provided clear evidence that **skip connections are essential** for retinal vessel segmentation, while **encoder and decoder complexity can be reduced** without significant performance loss. This is visible in the simplified encoder–decoder variant, where skip connections are preserved, reflecting their importance for the model.

Additionally, simplifying the encoder and decoder (reducing from DoubleConv to SingleConv) did not harm performance and in some cases improved it. This suggests the baseline U-Net is overparameterized for this task. Without regularization, the baseline U-Net reaches a validation Dice of 0.8683 and Focal Tversky of 0.6565, while the simple encoder–decoder achieved slightly lower Dice score (0.8521) but a higher Focal Tversky of 0.8227. When data augmentation and dropout were included, along with training using Focal Tversky loss. We found that the difference between the basic model and the simplified model increased. The regularized baseline reaches a validation Dice of 0.8272 and Focal Tversky of 0.8611, whereas the regularized simple encoder–decoder achieves a comparable Dice of 0.8320 and a higher Focal Tversky of 0.9061.

Research of Swin U-Net shows that transformer-based segmentation models generally need to be pre-trained or trained on the big datasets to have an influence. In the paper the authors **pre-trained** Swin U-Net on **ImageNet**.[4]. In our use case, the limited amount of images (480 samples after partitioning) limits the ability of the transformer network to train the most relevant patters to detect all the vessels while reducing the noise. The results show that the most important parameters in our specific use case are the window size and patch size. The reason is that the vessels are with various width and are thin. The smaller patches (2) proved to be better to represent thin vessels. The window size affects attention to tokens: a smaller size, like two, focuses on neighbors, while a larger size, like eight, looks at related features. In our case, we need to capture edges and texture, a smaller window size is more suitable for our needs.

The Attention U-Net with ASPP configuration showed stable results between Dice 0.8442 and Focal Tversky 0.8657, it would be worth it to consider further exploration.

5.2 Loss Function Trade-offs

The choice between Dice Loss and Focal Tversky Loss involves a fundamental trade-off. **Dice Loss** optimises overall segmentation quality (precision-recall balance) and is more robust for general-purpose vessel segmentation. It achieves higher Dice scores but may miss thin vessels. **Focal Tversky Loss** prioritises recall (detecting all vessels), making it superior for thin vessel detection. However, this comes at the cost of slightly lower Dice scores due to increased false positives.

For clinical applications where missing vessels could lead to misdiagnosis (e.g., diabetic retinopathy screening), **higher recall is preferable even at the expense of some false positives**. Conversely, if precise quantification of vessel morphology is required (e.g., measuring vessel caliber), minimising false positives (higher precision) may be more important, thus the use of Focal Tversky Loss is better in our test case.

We should also consider combination of Dice and other loss function to gather the strenghts of each one in a combined final loss function.

5.3 Overfitting and Regularization

The baseline U-Net demonstrated clear signs of overfitting. Low Focal Tversky scores (0.6565) despite high Dice (0.8683), indicating specialization to thick vessels and background. Large performance gap compared to simple models on Focal Tversky metric. Significant improvement ($0.6565 \rightarrow 0.8621$) after introducing augmentation and dropout.

This overfitting likely stems from insufficient training data diversity. The FIVES dataset, while large (800 images, of which we used 480 for training), may not cover the full variation in vessel morphology, lighting conditions, and pathological features. Data augmentation effectively increases dataset diversity, while dropout prevents co-adaptation of features, forcing the network to learn more robust representations.

The simple encoder-decoder model's superior Focal Tversky performance (even before regularization) suggests that **reducing model capacity is an effective overfitting mitigation strategy** for this task. This aligns with the principle of Occam's Razor: simpler models generalize better when data is limited.

5.4 Limitations

Our study has several limitations. First, it was evaluated on a single dataset FIVES, so its generalization to other retinal vessel segmentation datasets such as DRIVE, STARE, or CHASE-DB1 remains unknown. Second, to reduce computational cost we downsampled the images to 512×512, which may have removed fine vessel details present in the original 2048×2048 images. Third, the hyperparameters of the Focal Tversky loss (α, β, γ) were not exhaustively tuned, and the optimal values may differ from those used in our experiments. Fourth, we did not perform a systematic comparison to related work methods. Instead, we focused on establishing a strong baseline, so benchmarking against published approaches is still needed. Finally, our experiments were constrained by computational resources, which may have limited the extent of model and hyperparameter exploration.

5.5 What Worked and What Didn't

The benefits are that simplifying the encoder and decoder helped reduce overfitting while largely maintaining segmentation performance. Data augmentation and dropout acted as effective regularization strategies, improving generalization. The Focal Tversky metric was particularly useful for identifying cases where the model overfit to thick vessels and ignored thin ones. In addition, systematic ablation studies provided clear insight into the importance of individual architectural components.

The costs are that removing skip connections caused severe performance degradation, confirming their central role in U-Net–style architectures. Keeping the baseline model highly complex without sufficient regularization led to noticeable overfitting. Finally, relying solely on Dice loss proved insufficient, as it masked failures on thin vessel detection. These issues only became apparent once we evaluated models using the Focal Tversky metric.

Regarding the Swin U-Net, the architecture did not yield the performance improvements expected from transformer-based models. As noted in the architectural insights, the Swin U-Net consistently underperformed compared to the simpler CNN-based models in our specific setup. This highlights that simply replacing convolutions with transformers is not a guaranteed improvement, particularly when training data is limited and the model is not initialized with pre-trained weights from larger datasets.

6 Conclusion

This study systematically investigated architectural modifications and loss function choices for retinal vessel segmentation using U-Net. Rather than proposing a novel architecture that outperforms the related work, our work validates the efficiency of streamlined models in resource-constrained scenarios. Through comprehensive ablation studies, we demonstrated several key findings. At first, **model simplification reduces overfitting**. The simple encoder–decoder model achieves competitive DICE scores (0.8516) and superior Focal Tversky scores (0.9056) compared to the baseline, indicating better generalisation to thin vessels. Second, **regularisation is effective**. Data augmentation and dropout dramatically improved baseline model performance (Focal Tversky from 0.6565 to 0.8621), closing the gap with the simple model. Moreover, **loss function choice matters**. Dice Loss and Focal Tversky Loss optimise different aspects of segmentation quality, so the appropriate choice depends on clinical priorities, thus, balancing overall performance versus maximising recall of challenging thin vessels. Finally, **Transformer efficacy is data-dependent**. Our results indicate that **without large-scale pre-training or massive datasets**, Transformer-based architectures like Swin U-Net are prone to overfitting and fail to outperform efficiently designed convolutional baselines in low-data regimes.

Future work could focus on properly leveraging the Swin U-Net architecture by utilising pre-training on large-scale datasets such as ImageNet before fine-tuning on medical imaging data or using other methods to increase dataset size.

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